WEEK 2 CYBERSECURITY CHEAT SHEET

Network Traffic Analysis • Nmap • Windows Logging

1. TCP FLAGS — QUICK MEMORY

SYN = start connection | **ACK** = acknowledge receipt | **FIN** = orderly close

RST = reset/reject connection | **PSH** = prompt delivery to application | **URG** = urgent pointer significant

Handshake: SYN → SYN-ACK → ACK | **Close:** FIN/ACK exchanges commonly | **RST:** reset/aborted connection

2. PROTOCOL BEHAVIOUR — WHAT TO NOTICE

DNS: expected domain queries vs repeated/unusual queries or unexpected DNS server.

HTTP: normal web requests/responses vs unexpected downloads, suspicious destinations, unusual methods, clear-text credentials.

HTTPS/TLS: expected encrypted sessions vs unusual destinations, certificate anomalies, repeated failed handshakes. *Encryption alone is not suspicious.*

ARP: local IP-to-MAC resolution vs unexpected replies or duplicate mappings (possible spoofing).

ICMP: ping/errors vs high-volume probes, sweeps, or unexpected tunnelling patterns.

SMB: authorised file/printer sharing vs unexpected connections, lateral-movement patterns, repeated authentication failures.

3. WIRESHARK DISPLAY FILTERS

Protocol: `tcp` · `udp` · `icmp` · `arp` · `dns` · `http` · `tls`

`tcp.port == 80` = HTTP port | `tcp.port == 443` = HTTPS/TLS port

`ip.addr == 192.168.1.10` = traffic to/from host | `ip.src == 192.168.1.10` = source | `ip.dst == 192.168.1.10` = destination

`tcp.flags.syn == 1 && tcp.flags.ack == 0` = initial SYN probes

`tcp.flags.reset == 1` = RST packets | `dns.flags.response == 0` = DNS queries

`http.request.method == "GET"` = HTTP GET requests | `tcp.analysis.retransmission` = retransmissions

`tcp.stream eq 0` = isolate stream 0 (change number as needed). Display filters apply after capture/load; capture-filter syntax differs.

4. NMAP COMMANDS — AUTHORISED LABS ONLY

`nmap -sn <target-or-subnet>` = host discovery (no standard port scan)

`nmap -sS <target-IP>` = TCP SYN scan | `nmap -sT <target-IP>` = TCP connect scan

`nmap -sU <target-IP>` = UDP scan | `nmap -sV <target-IP>` = service/version detection

`nmap -O <target-IP>` = OS detection attempt | `nmap -A <target-IP>` = multiple advanced detections; noisier

`nmap -sS -oN scan.txt -oX scan.xml <target-IP>` = save output in normal and XML formats

Response clues: SYN-ACK commonly means open; RST commonly means closed; no UDP response may mean open|filtered. Firewalls/network conditions affect results.

5. WINDOWS LOGS & SECURITY EVENT IDs

Logs: Security = logons, account changes, privilege/audit activity; **System** = OS/service events; **Application** = application events.

4624 successful logon | **4625** failed logon | **4634** logoff | **4648** explicit credentials

4672 special privileges assigned | **4688** new process created | **4697** service installed (if audited)

4720 user created | **4722** user enabled | **4724** password reset attempt | **4732** added to local security-enabled group

4768 Kerberos TGT requested | **4769** Kerberos service ticket requested | **4776** NTLM credentials validated | **1102** Security audit log cleared

Event availability depends on audit policy, Windows version, retention, and where the event is generated. IDs 4768/4769 are generally reviewed on domain controllers.

6. BASELINE + SOC REMINDERS

Baseline: documented normal activity for a system, user, application or network. Track expected hosts, ports, DNS destinations, login times, services, processes and event volumes.

Investigate deviations: unusual ports/destinations, login times, process launches or event volumes. Document approved maintenance/scanners to reduce false positives.

Remember: a deviation is a lead to investigate, not proof of an attack. Use Nmap only on authorised systems; record target scope, time, command and actual result.

Prepared from the Week 2 report • Aayat Ansari • DG Interns Hub Cyber Security Internship